R Practice 5 – The Tidyverse

This assignment is just for your own practice and does not need to be submitted. If you have any questions, please feel free to reach out to me, come to office hours, or schedule another a time to meet with me.

In this assignment, we are focusing on using tidyverse functions. You should use pipes when more than two functions are involved and the functions **mutate()**, **select()**, **filter()**, **arrange()**, **summarize()**, and and **group_by()** whenever you can. Open an R Script in RStudio and complete each of the following:

1. **Flights.** We will be dealing with the flights data set in the **nycflights13** package. This data set contains on-time data for all flights that departed NYC (i.e. JFK, LGA or EWR) in 2013. More info about the data set can be found [here].

    a. Install and load the **nycflights13** package. Print the **flights** data set by typing its name. Also type **?flights** to get more info about this data set.

    b. Using the appropriate tidyverse function, create a new variable in the **flights** called **speed** that is taken by dividing **distance** by **air_time** and multiplying that by 60 (since **air_time** is in minutes). This will give you the average speed of the plane during the flight.

    c. Find the average speed for each of the origin airports in this data set. You should use the **group_by()** and **summarize()** functions here. Note: there are some NA values here, so you should put **na.rm = T** in the **mean()** function when you use it.

    d. Find the average speed for each of the carriers in this data set and arrange them from greatest to least. Which airline had the fastest average speed? Which airline had the slowest average speed? The **arrange()** function is helpful for answering those questions.

    e. Now find the average, max, and min arrival delay for each month of the year. This can be found in the **arr_delay** variable where negative times indicate early arrivals. Remember to put **na.rm = T** in the **mean**, **min**, and **max** functions. Which month has the lowest average delay time? Which month had the lowest max delay time?

    f. Find the average arrival delay for the “big four” carriers: American Airlines (AA), Southwest Airlines (WN), Delta Airlines (DL), and United Airlines (UA), and in each month. You can put two variables in the **group_by()** function and you will need the **filter()** function. Print all 48 rows by piping this into **print(n = 48)**. Does any airline have a positive average delay time in every month?

g. Create a tibble called **slc_flights** that only contains the rows with a destination of Salt Lake City (which is the SLC airport). Then reduce this down so that only the variables **carrier**, **origin**, **air_time**, and **speed** are selected. How many flights flew from New York City to Salt Lake airport in 2013?

h. Now filter this down even more so that all the rows with missing **air_time** values are removed. You can use the **is.na()** function to check if the value is missing.

i. Using the tibble without any missing values, find the min, first quartile, median, third quartile, and max amount of time it took to get to SLC in hours. Then get the min, first quartile, median, third quartile, and max of the speed. You can get the quartiles using the **quantile()** function. For example, **quantile(x, 0.25)** will give the first quartile of the vector **x**.